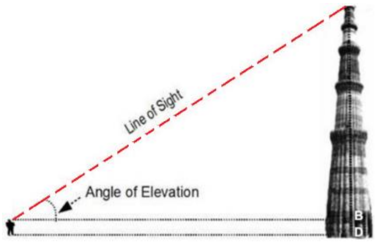
Applications of Differentiation

A. Related rates
B. Properties of curves (extrema, concavity etc)
C. Optimization
D. L'Hospital's rule and Indeterminate forms
E. Newton's method in roots finding



A1. Related rates (review of relevant concepts)

Problems on related rates is the application of implicit differentiation. The technique of implicit differentiation allows us to find the derivative of y (the targeted function) with respect to t (the independent variable) while the chain rule must be used in this technique. Before entering the subject of related rates, we briefly review the relevant concepts.

Rev-1: Implicit vs Explicit
A function can be explicit or implicit:

- Explicit: "y = some function of t ". When we know t we can calculate y directly.
  E.g. $y = t\cos(t^2+1)$

- Implicit: "some function of y and t equals something else". Knowing t does not lead directly to y.
  E.g. $y^2+t^2-2t+3y=5;\quad ty+\sin(ty)=1$

Rev-2: Composite function
When a function (f) is applied to the result of another function (g), then a composite function $f(g(t))$ is generated. The composite function $f(g(t))$ is often denoted as $(f\circ g)(t)$ and is read as "f of g of t". The function $g(t)$ is called an inner function and the function $f(t)$ is called an outer function. In general, $(f\circ g)(t)$ is not the same as $(g\circ f)(t)$. Whenever the argument of a function is not an independent variable, but another function, then this function is a composite function.

E.g. Let $f(t)=\sin(t)$; $g(t)=t^3+1$
- $f(g(t)) = \sin(g(t)) = \sin(t^3+1)$
- $g(f(t)) = [f(t)]^3+1 = 1+\sin^3 t$

Rev-3: The chain rule
The chain rule provides us a technique for finding the derivative of composite functions. The number of differentiation steps needed equals the number of functions that make up the composition. The chain rule states formally that

$D[f(g(t))] = f'(g(t))g'(t)$ where $D[\cdot]$ denotes "the derivative of".

Or, $D[f(g(t))] = \frac{df}{dt} = \frac{df}{dg}\cdot\frac{dg}{dt}$

$$\begin{gathered} t \\ \downarrow \\ g(t)=g \\ \downarrow \\ f(g)=f \end{gathered}$$

Or, in words, the derivative of a composition is the derivative of the outer w.r.t. the inner (the inside staying the same) TIMES the derivative of the inner.

E.g.1 $\frac{d}{dt}\sin(t^3+t)$

$$\frac{df}{dt} = \frac{df}{dg}\cdot\frac{dg}{dt}$$
$$= \cos(g)\ (3t^2+1)$$
$$= \cos(t^3+t)\ (3t^2+1)$$

$$\begin{gathered} t \\ \downarrow \\ g(t)=t^3+t \\ \downarrow \\ f(g)=\sin(g) \end{gathered}$$

It may involve more than two differentiation steps. Remember, the number of differentiation steps needed is equal to the number of functions that make up the composition.

E.g.2 $\frac{d}{dt}\sin^2(t^2+1) = \frac{d}{dt}[\sin(t^2+1)]^2$

$\frac{df}{dt} = \frac{df}{dq}\cdot\frac{dq}{dp}\cdot\frac{dp}{dt}$ [3 differentiation steps

$= 2q\cdot\cos(p)\cdot(2t)$

$= 2\sin(t^2+1)\cdot\cos(t^2+1)\cdot(2t)$

$$\begin{array}{c} t \\ \downarrow \\ p(t) = t^2+1 \\ \downarrow \\ q(p) = \sin(p) \\ \downarrow \\ f(q) = q^2 \end{array}$$

We often need to use C-R together with other differentiation rules (such as the product rule and quotient rule) to differentiate more complicated functions.

E.g.3 $\frac{d}{dt}t^3\sin(t^2+1)$

$\frac{d}{dt}g(t)f(t) = g\frac{df}{dt} + f\frac{dg}{dt}$ (the product rule)

[where $g(t) = t^3$; $f(t) = f(p(t))$

$= g\left[\frac{df}{dp}\cdot\frac{dp}{dt}\right] + f\frac{dg}{dt}$ (C-R is necessary for $\frac{df}{dt}$)

$= t^3[\cos(t^2+1)\cdot(2t)] + \sin(t^2+1)\cdot(3t^2)$

$$\begin{array}{c} t \\ \downarrow \\ p(t) = t^2+1 \\ \downarrow \\ f(p) = \sin(p) \end{array}$$

E.g.4 $D[\cos(t^3\sin(t^2))]$

$= \frac{d}{dt}f\big(g(t)\big)$ [where $f(\cdot) = \cos(\cdot)$ and $g(t) = t^3\sin(t^2)$

$= \frac{df}{dg}\cdot\frac{dg}{dt}$, but $\frac{dg}{dt}$ requires the product rule

$= -\sin(g)\cdot[t^3\cos(t^2)\cdot 2t + \sin(t^2)\,(3t^2)]$

$= -\sin(t^3\sin(t^2))\cdot[2t^4\cos(t^2) + 3t^2\sin(t^2)]$

$$\begin{array}{c} t \\ \downarrow \\ g(t) = t^3\,sin(t^2) \\ \downarrow \\ f(g) = \cos(g) \end{array}$$

Ex-1 Find $\frac{dz}{dt}$ for $z = \sin(t^2)\cos(t^3)$.

Ans: $2t\cos[t^2]\cos[t^3] - 3t^2\sin[t^2]\sin[t^3]$

Ex-2 Find $\frac{dz}{dt}$ for $z = \sin[\cos(t^3)\tan(t^2)]$

Ans: $\cos(\cos[t^3]\tan[t^2])\ \{2t\cos[t^3]\,(\sec[t^2])^2 - 3t^2\sin[t^3]\tan[t^2]\}$

Rev-4: implicit differentiation

This is a technique to differentiate a function that is defined implicitly. Remember, this technique will always involve chain rule. For example, suppose y is defined implicitly by $y^2+t^3-y^3+6=3y$ and we want to find dy/dt. For this technique, first we differentiate the equation with respect to t (term-wise), then we collect all the dy/dt on one side and finally we solve for dy/dt. Very often, dy/dt is solved as a function of t and y, but not of t alone.

E.g.1: Find dy/dt for $y^2+t^3-y^3+6=3y$

Step-1: d/dt term-wise

$\frac{d}{dt}(y^2) + \frac{d}{dt}(t^3) - \frac{d}{dt}(y^3) + \frac{d}{dt}(6) = \frac{d}{dt}(3y)$

Be mindful that functions of y are actually composite functions. For instance, $f = y^2$ should be regarded as $f = f(y(t)) = (y(t))^2$ where $f(y) = y^2$. Another example, $g = \sin(y)$ should be regarded as $g = g(y(t)) = \sin(y(t))$. Here, the argument of g is the function y, instead of the independent variable t. Therefore, d/dt of y-functions involve the chain-rule.

$$t \to y(t) = y \to f(y) = y^2$$

$$t \to y(t) = y \to g(y) = \sin(y)$$

For better understanding, we rename the y-functions such that $y^2 = f(y)$ and $y^3 = h(y)$. So now,

$\frac{d}{dt}(y^2) + \frac{d}{dt}(t^3) - \frac{d}{dt}(y^3) + \frac{d}{dt}(6) = \frac{d}{dt}(3y)$ becomes

$\frac{d}{dt}(f(y)) + \frac{d}{dt}(t^3) - \frac{d}{dt}(h(y)) + \frac{d}{dt}(6) = \frac{d}{dt}(3y)$. Applying C-R gives

$$\frac{df}{dy} \cdot \frac{dy}{dt} + \frac{d}{dt}(t^3) - \frac{dh}{dy} \cdot \frac{dy}{dt} + \frac{d}{dt}(6) = \frac{d}{dt}(3y)$$

$$2y\frac{dy}{dt} + 3t^2 - 3y^2\frac{dy}{dt} + 0 = 3\frac{dy}{dt}$$

Step-2: collect all the dy/dt on one side

$$\frac{dy}{dt}[3 - 2y + 3y^2] = 3t^2$$

Step-3: solve for the dy/dt

$$\frac{dy}{dt} = \frac{3t^2}{3-2y+3y^2}$$

E.g.2: Find dy/dt if $t^2y^3 - ty = 9$

Note that besides C-R, we need to use the production rule (for the first two terms) here.

- Term-wise d/dt: $2t\,y^3 + t^2\left(3y^2\frac{dy}{dt}\right) - (1)y - t\left(\frac{dy}{dt}\right) = 0$
- Collecting dy/dt: $\frac{dy}{dt}[3t^2y^2 - t] = y - 2ty^3$
- Solving for dy/dt: $\frac{dy}{dt} = \frac{y-2ty^3}{3t^2y^2-t}$

E.g.3: Find $\frac{dy}{dt}$ for which y is defined by $\sin y + ty^3 = \cos t$

$$\cos y\frac{dy}{dt} + \left(t \cdot 3y^2\frac{dy}{dt} + y^3\right) = -\sin t$$

$$\frac{dy}{dt}[\cos y + 3ty^2] = -\sin t - y^3$$

$$\frac{dy}{dt} = \frac{-\sin t - y^3}{\cos y + 3ty^2}$$

A2. Finding related rates

1) It is common for variables to be functions of time. In certain problems, a few quantities / variables are related by an equation meanwhile all the variables are functions of a time (t) and are changing at a certain rate. Then, the relation of their rates of change may be found by differentiating (implicitly) both sides of the equation. This kind of problem is said to involve related rates.

Eg1. $x = x(t)$, $y = y(t)$, given $y = xy^2 + 1$, then $\frac{dy}{dt} =? \frac{dx}{dt}$

$$\frac{dy}{dt} = \frac{dx}{dt}y^2 + 2xy\frac{dy}{dt} \Rightarrow \frac{dy}{dt} = \left(\frac{dx}{dt}y^2\right)/(1 - 2xy)$$

If the x-y relation is known, can we find the $\frac{dx}{dt}$-$\frac{dy}{dt}$ relation?

Eg2. When water is drained out of a conical tank, the volume V, the radius r, and the height h of the water level are all functions of time t.

Knowing that these variables are related by the equation

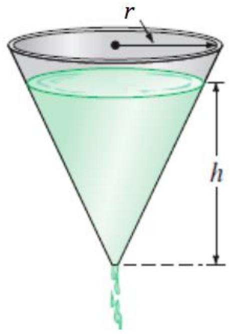
$$V = \frac{\pi}{3} r^2 h$$ Original equation

you can differentiate implicitly with respect to t to obtain the **related-rate** equation

$$\frac{d}{dt}[V] = \frac{d}{dt}\left[\frac{\pi}{3} r^2 h\right]$$

$$\frac{dV}{dt} = \frac{\pi}{3}\left[r^2 \frac{dh}{dt} + h\left(2r\frac{dr}{dt}\right)\right]$$ Differentiate with respect to t.

$$= \frac{\pi}{3}\left(r^2 \frac{dh}{dt} + 2rh\frac{dr}{dt}\right).$$



From this equation, you can see that the rate of change of V is related to the rates of change of both h and r.

2) Typically, in related-rates problems, the values of a few time varying quantities at some instant are given, together with all their time rates of change but one. The problem is usually to find the time rate of change of the one that is not given, at some instant specified in the problem.

Eg3.

Suppose $(x(t), y(t))$ are the coordinates at time t of a point moving around the circle having equation $x^2 + y^2 = 25$. At a certain instant, given $x = 3, y = 4$ and $dx/dt = 12$, $dy/dt = ?$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$2 \cdot 3 \cdot 12 + 2 \cdot 4 \cdot \frac{dy}{dt} = 0$$

So, $dy/dt = -9$

Notice here that the expressions of x and y as functions of t are both not given.

3) Solving related rate problems will involve the implicit differentiation (and hence the C-R). It can be regarded as the application the implicit differentiation.

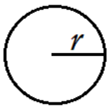
4) Guidelines for solving related-rate problems
   (i). Identify all variables involved and variables to be determined. Sketch a diagram and label the variables involved.
   (ii). Record the values of the variables and their rates of change, as given in the problem.
   (iii). Write an equation involving the variables. If the equation is not given or given indirectly, use the diagram to determine the equation that relates the variables.
   (iv). Implicitly differentiate both sides of the equation with respect to time t (C-R is necessary).
   (v). Substitute the given numerical data in the resulting equation, and then solve for the unknown.

It is common for a related-rates problem to contain insufficient information to express the time varying variables as functions of t. Geometric properties such as similar triangles, Pythagorean theorem are always useful.

Example-1 涟漪

When a small stone is dropped into a lake, circular ripples are spreading over the surface of the water, with the radius of each circle increasing at the rate of 3/2 ft per second. Find the rate of change of the area inside the circle formed by a ripple at the instant the radius is 4 ft.

$A = \pi r^2$ and both A, r vary with time



$$\frac{d}{dt}(A) = \frac{d}{dt}(\pi r^2)$$

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt}$$

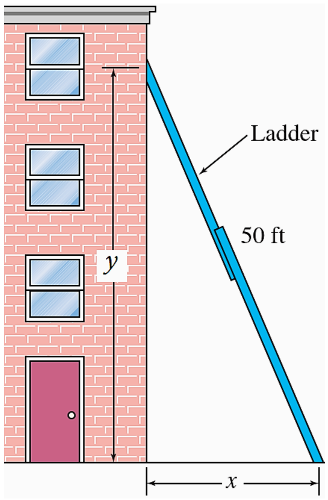
Given $\frac{dr}{dt} = \frac{3}{2}$, $r = 4$, ask for $\frac{dA}{dt}$?

$$\frac{dA}{dt} = 2\pi \cdot 4 \cdot \frac{3}{2}$$

$$\frac{dA}{dt} = 12\pi \approx 37.7 \text{ ft}^2 \text{ per second.}$$

Example-2

A 50-ft ladder is placed against a large building. The base of the ladder is resting on an oil spill, and it slips at the rate of 3 ft per minute. Find the rate of change of the height of the top of the ladder above the ground at the instant when the base of the ladder is 30 ft from the base of the building.



$\frac{dx}{dt} = 3, x = 30, \frac{dy}{dt} = ?$

Pythagorean theorem: $x^2 + y^2 = 50^2$

$\frac{d}{dt}$ implicitly: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

When $x = 30$, $30^2 + y^2 = 50^2 \implies y = 40$

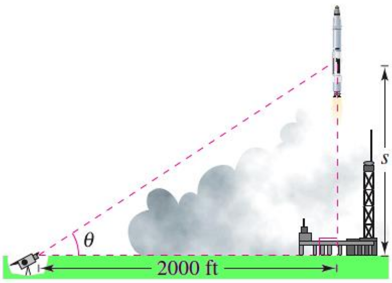
So, $\frac{dx}{dt} = 3, x = 30, y = 40, \frac{dy}{dt} = ?$

$2(30)3 + 2(40)\frac{dy}{dt} = 0, \frac{dy}{dt} = -\frac{9}{4} = -2.25$ (minus sign shows decreasing)

The top of the ladder is sliding down the building at the rate of 2.25 ft per minute.

Example-3

A television camera at ground level is filming the lift-off of a rocket that is rising vertically according to the position equation $s = 50t^2$ where is s measured in feet and t is measured in seconds. The camera is 2000 feet from the launch pad. Find the rate of change in the angle of elevation of the camera at 10 seconds after lift-off.



$s, \theta,\ t = 10, \frac{d\theta}{dt} = ?$

The s, θ relationship is not given directly, $\frac{ds}{dt}$ is not given directly.

When $t = 10, s = ?\ \ \theta = ?$ not given directly.

When $t = 10$, the height s of the rocket is
$s = 50t^2 = 50(10)^2 = 5000$ feet.

Given: $ds/dt = 100t =$ velocity of rocket (indirectly from $s = 50t^2$)

Find: $d\theta/dt$ when $t = 10$ and $s = 5000$

s and θ by the equation $\tan\theta = s/2000$

$$\tan\theta = \frac{s}{2000}$$

$$(\sec^2\theta)\frac{d\theta}{dt} = \frac{1}{2000}\left(\frac{ds}{dt}\right) \qquad \text{Differentiate with respect to } t.$$

$$\frac{d\theta}{dt} = \cos^2\theta\,\frac{100t}{2000} \qquad \text{Substitute } 100t \text{ for } \frac{ds}{dt}.$$

$$= \left(\frac{2000}{\sqrt{s^2+2000^2}}\right)^2\frac{100t}{2000} \qquad \cos\theta = \frac{2000}{\sqrt{s^2+2000^2}}$$

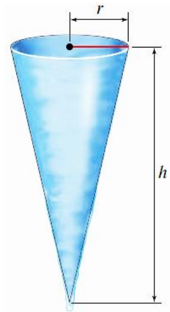
When $t = 10$ and $s = 5000$, you have

$$\frac{d\theta}{dt} = \frac{2000(100)(10)}{5000^2+2000^2} = \frac{2}{29} \text{ radian per second.}$$

So, when $t = 10$, θ is changing at a rate of $\frac{2}{29}$ radian per second.

Example-4 (more than 2 variables)

A cone-shaped icicle is dripping from the roof. The radius of the icicle is decreasing at a rate of 0.2 cm per hour, while the length is increasing at a rate of 0.8 cm per hour. If the icicle is currently 4 cm in radius and 20 cm long, at what rate is the volume of the icicle increasing (or decreasing)?



$$V = \frac{1}{3}\pi r^2 h,\ r = 4,\ \frac{dr}{dt} = -0.2,\ h = 20,\ \frac{dh}{dt} = 0.8,\ \frac{dV}{dt} = ?$$

$$\frac{dV}{dt} = \frac{1}{3}\pi\left[r^2\frac{dh}{dt} + (h)(2r)\frac{dr}{dt}\right]$$

$$\frac{dV}{dt} = \frac{1}{3}\pi[4^2(0.8)+(20)(8)(-0.2)]$$

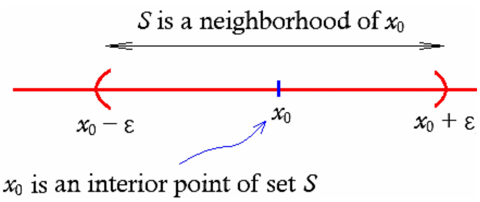
$$= \frac{1}{3}\pi(-19.2) \approx -20$$

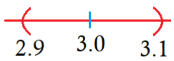
B. Properties of curves

B1. Definitions

1) Neighborhood of x_0, interior point x_0

A **neighborhood of a point x_0** means any set that contains an open interval containing x_0; i.e. an open interval of the form $(x_0 - \varepsilon, x_0 + \varepsilon)$, where $\varepsilon > 0$ is some small number. This constitutes set S, the set of points "near" x_0. We call a point x_0 an **interior point** of a set S if S is a **neighborhood** of x_0.





In words, a neighborhood is an interval that spreads from the central point to its neighboring areas.

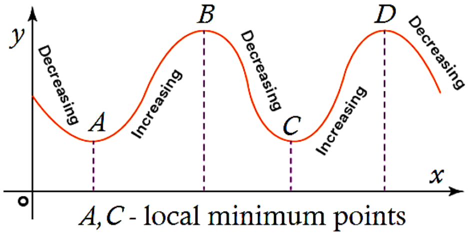
2) Increasing, decreasing functions, monotonic

A function f is **increasing** (resp., **decreasing** on an interval if for any two points $x_1 < x_2$ in this interval we have $f(x_1) < f(x_2)$ (resp. $f(x_1) > f(x_2)$). We say that $f(x)$ is increasing (resp. decreasing) at $x = c$ provided that $f(x)$ is increasing (decreasing) in a neighborhood of the point c. Geometrically, a function is increasing when, as x moves to the right, its graph moves up, and is decreasing when its graph moves down. A function that is increasing or decreasing on an interval I is called **monotonic** on I.

singular	plural
minimum, maximum	minima, maxima
extremum	extrema

3) Local/relative extrema

The graph of the function $f(x)$ is said to have a **local maximum** at $x = c$ if $f(c) > f(x)$ for all x in a neighborhood of c. We can say that f has a local maximum at $(c, f(c))$. Similarly, the graph has a **local minimum** at $x = c$ if $f(c) < f(x)$ on such an interval. Collectively, local maxima and minima are called **local extrema**. Geometrically, A local maximum point is a point at which the graph changes from increasing to decreasing; a local minimum point is a point at which the graph changes from decreasing to increasing.



A, C - local minimum points
B, D - local maximum points

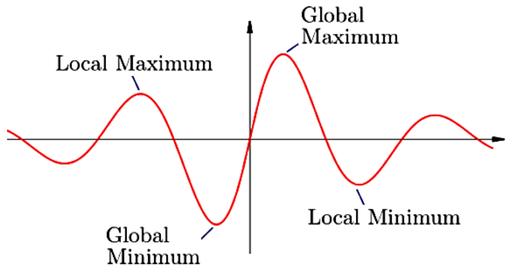
4) Global/absolute extrema

Let f be a function with domain D. Then f has an **absolute maximum** value on D at a point c if

$$f(c) \geq f(x) \quad \forall\, x \in D$$

and an **absolute minimum** value on D at c if

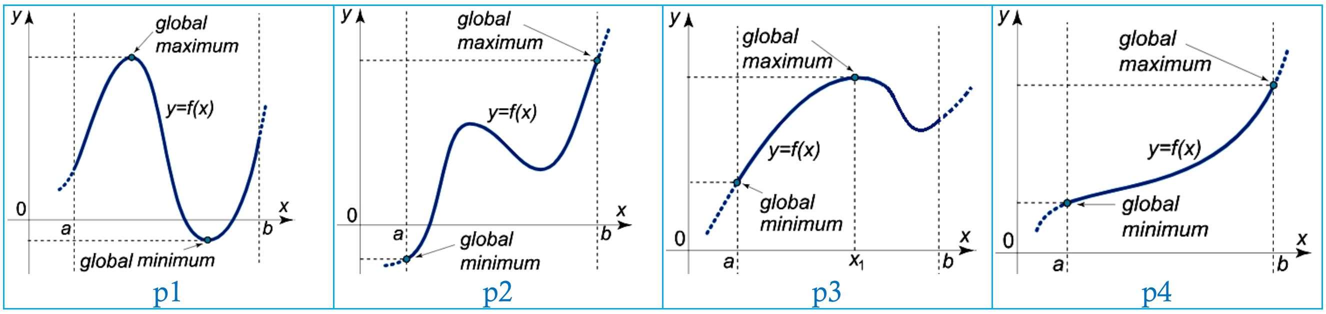
$$f(c) \leq f(x) \quad \forall\, x \in D$$



Absolute maximum and minimum values are collectively called **absolute extrema** (also called **global extrema**). They are the largest/smallest function values throughout the whole (targeted) domain.

5) Endpoint extrema

Absolute extrema can occur at interior points or endpoints of an interval. Absolute extrema that occur at the endpoints are called **endpoint extrema**. If f is continuous on a closed interval [a, b], then there must exist an absolute maximum value and an absolute minimum value of f in [a, b] (even though there are no critical points in $[a, b]$).



p1: Both the global extrema exist inside the interval $[a, b]$. (Interior-point extrema)

p2: Both the global extrema exist on the boundary of the interval $[a, b]$. They are end-point extrema.

p3: The global maximum exists inside whereas the global minimum exists on the boundary of the interval $[a, b]$. It is an end-point minimum.

p4: The global extrema exist on the boundary of the interval $[a, b]$ even though the function has no critical points on the interval $[a, b]$.

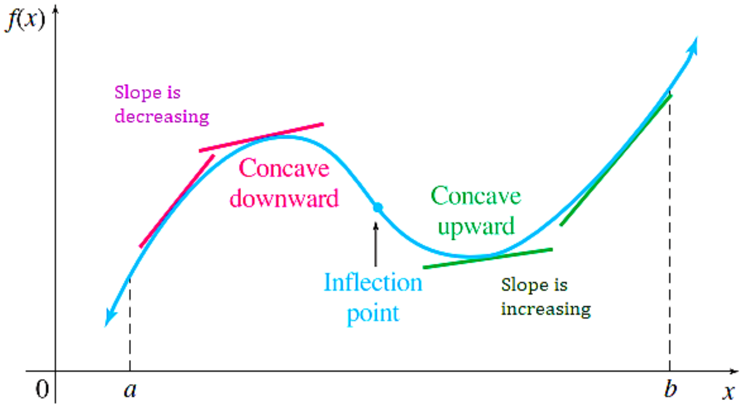
6) Concavity

The graph of a differentiable function $f(x)$ is

(a) **concave up** on an open interval I if its derivative f' is increasing on I

(b) **concave down** on an open interval I if its derivative f' is decreasing on I.

In colloquial terms, the graph of a function that is concave up is “smiling”, while the graph of a function that is concave down is “frowning”. 皱眉



7) Inflection point
A point where the graph of a function has a tangent line and where the concavity switches is a **point of inflection**. The switch may from upward to downward (or downward to upward).

8) Critical point
Let f be defined at c. If $f'(c) = 0$ or undefined (i.e. f is not differentiable at c) then c is a **critical point** of f.

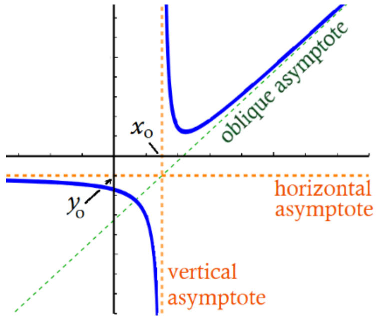
9) Stationary point
We call c a **stationary point** of f if $f'(c) = 0$.

10) Asymptote
Two curves are **asymptotic** if they approach each other “at infinity”; in particular, a line asymptotic to a curve is an **asymptote** to it. The graph $y = f(x)$ has

(a) a **horizontal asymptote** at $y = y_0$ if either $\lim_{x\to\infty} f(x) = y_0$ or $\lim_{x\to-\infty} f(x) = y_0$

(b) has a **vertical asymptote** at $x = x_0$ if $f(x)$ diverges to $\pm\infty$ as $x \to x_0$ from either the right or the left.

* Notice that asymptote can be a line in any direction besides vertical and horizontal. It is called the slanting asymptote (or oblique asymptote).



B2. Theorems and tests

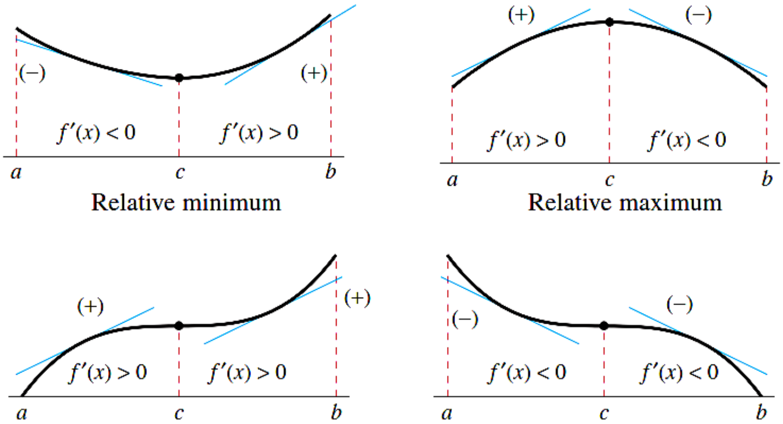
1) First derivative test for **monotonic** functions
Suppose that f is continuous on $[a, b]$ and differentiable on (a, b).
(a) If $f'(x) > 0$ at each point $x \in (a, b)$, then f is **increasing** on $[a, b]$.
(b) If $f'(x) < 0$ at each point $x \in (a, b)$, then f is **decreasing** on [a, b].
(c) If $f'(x) = 0$ at each point $x \in (a, b)$, then f is **constant** on [a, b] (and hence horizontal).

2) Second derivative test for **concavity**
Let f be twice differentiable on an open interval.
(a) If $f''(x) > 0$ for every value of x in the open interval, then f is **concave up** on that interval.
(b) If $f''(x) < 0$ for every value of x in the open interval, then f is **concave down** on that interval.

3) Second derivative test for **local extrema**
Suppose f'' is continuous on an open interval that contains $x = c$. If $f'(c) = 0$, and
(a) $f''(c) < 0$, then f has a **local maximum** at $x = c$.
(b) $f''(c) > 0$, then f has a **local minimum** at $x = c$.

(c) $f''(c) = 0$, then the test is inconclusive. The function f may have a local maximum, a local minimum, or neither (or an inflection point). In such cases, you can use the First derivative test.

4) First Derivative Test for **local extrema**
Local extrema must occur at critical points, but not all critical points are local extrema.
Suppose that f is continuous at a critical point c. When passing through point c from its left neighbor to its right neighbor, if
(a) $f'(x)$ changes from negative to positive, then f has a local minimum at c.
(b) $f'(x)$ changes from positive to negative, then f has a local maximum at c.
(c) $f'(x)$ has the same sign, then f does not has a local extremum at c.



Neither relative minimum nor relative maximum

5) Second derivative test for **inflection point**
If $(c, f(c))$ is an inflection point of the graph of f, then either $f''(c) = 0$ or f'' does not exist at $x = c$.
Suppose that $f''(c) = 0$ or $\nexists$. Extending from the left neighbor of c to its right neighbor, if $f''(x)$ switches sign, then c is an inflection point.

6) Third derivative test for **inflection point**
If $f''(c) = 0$ and $f'''(c) \neq 0$, then c is an inflection point of the function $f(x)$.

7) The Extreme Value Theorem
If a function f is continuous on a finite closed interval $[a, b]$, then f has both an **absolute maximum** and an **absolute minimum** on $[a, b]$.

8) If f has an **absolute extremum** on an open interval (a, b), then it must occur at a **critical point** of f, otherwise, it must occur at the endpoints (at a or b).

9) Finding absolute extrema on a finite closed interval $[a, b]$
Step 1. Find the critical points of f in (a, b).
Step 2. Evaluate f at all the critical points <u>and also at the endpoints</u> a and b.
Step 3. The largest of the values in *Step 2* is the absolute maximum value of f on $[a, b]$ and the smallest value is the absolute minimum.

Example-5

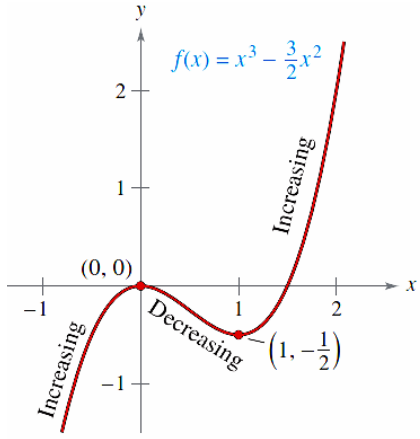
Consider the function $f(x) = x^3 - \frac{3}{2}x^2$.

(a) Find the open interval on which $f(x)$ is increasing and decreasing.

(b) Find the open interval on which $f(x)$ is concave up and down.

(c) Identify all the local or global extrema using various tests.

(d) Identify all the inflection points using various tests.



$f'(x) = 3x^2 - 3x;\ \ f''(x) = 6x - 3;\ \ f'''(x) = 6$

Note: $\begin{cases} f' = 0 \longrightarrow \text{potential candidated of local Min/Max} \\ f'' = 0 \longrightarrow \text{potential candidated of inflection point} \end{cases}$

(a) Critical points:

$f'(x) = 0$ when $x = 0, 1 \Longrightarrow$ critical points

Interval	$-\infty < x < 0$	$0 < x < 1$	$1 < x < \infty$
Test Value	$x = -1$	$x = \frac{1}{2}$	$x = 2$
Sign of $f'(x)$	$f'(-1) = 6 > 0$	$f'(\frac{1}{2}) = -\frac{3}{4} < 0$	$f'(2) = 6 > 0$
Conclusion	Increasing	Decreasing	Increasing

So, f is increasing on the intervals $(-\infty, 0)$ and $(1, \infty)$ while decreasing on the interval (0,1).

(b) $f''(x) = 6x - 3 = 0$ when $x = 1/2$

For interval $x < 1/2$, test value $x = 0$: $f''(0) = -3 \Longrightarrow$ concave down

For interval $x > 1/2$, test value $x = 1$: $f''(1) = 3 \Longrightarrow$ concave up

So, $f(x)$ concaves down for $x < 1/2$, and concaves up for $x > 1/2$.

(c) Critical points: $x = 0, 1$

▶ First derivative test, $f'(x) = 3x^2 - 3x$

	$0 - \varepsilon$	0	$0 + \varepsilon$		$1 - \varepsilon$	1	$1 + \varepsilon$
Test value	--0.1	0	0.1		0.9	1	1.1
$f'(x)$	0.33	0	-0.27		-0.27	0	0.33
monotonicity	increase	level	decrease		decrease	level	increase
shape	↗	⟶	↘		↘	⟶	↗
inference		max				min	

▶ Second derivative test, $f''(x) = 6x - 3$

Critical points: $x = 0, 1$

$f'(0) = 0,\ f''(0) = -3 < 0 \ \Longrightarrow$ Maximum

$f'(1) = 0,\ f''(1) = \ 3\ > 0 \ \Longrightarrow$ Minimum

(d) Potential inflection point, $f''(x) = 6x - 3 = 0$ when $x = 1/2$

▶ Second derivative test

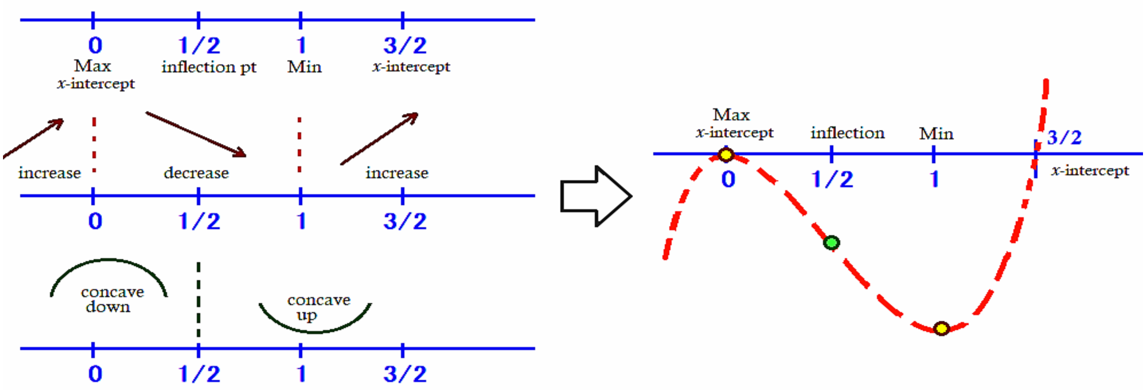
	$1/2 - \varepsilon$	$1/2$	$1/2 + \varepsilon$
Test value	$1/4$	½=2/4	$3/4$
$f''(x)$	$-3/2$	0	$3/2$
concavity	down		up
shape	$\frown$		$\smile$
inference		inflection point	

▶ Third derivative test, $f''(x) = 6x - 3;\ \ f'''(x) = 6$

Potential inflection point $x = 1/2$

$f''(1/2) = 0,\ f'''(1/2) = 6 \neq 0 \implies$ inflection point

A collective visualization after gathering all information.



Example-6 (Extrema on a closed interval, endpoint extrema)

For several weeks, the traffic department has been recording the speed of highway traffic flowing past a certain downtown exit. The data suggest that between 1:00 and 6:00 pm on a normal weekday, the speed of the traffic at the exit is approximately $S(t) = t^3 - 10.5t^2 + 30t + 20$ km/h, where t is the number of hours past noon. At what time between 1:00 and 6:00 pm is the traffic moving the fastest, and at what time is it moving the slowest?

The goal is to find the absolute maximum and absolute minimum of the function $S(t)$ on the interval $1 \leqslant t \leqslant 6$. From the derivative

$$S'(t) = 3t^2 - 21t + 30 = 3(t^2 - 7t + 10) = 3(t - 2)(t - 5)$$

The critical points are at $t = 2$ and $t = 5$, both of which lie in the interval $1 \leqslant t \leqslant 6$..

Compute $S(t)$ for the critical values and also at the endpoints $t = 1$ and $t = 6$ give

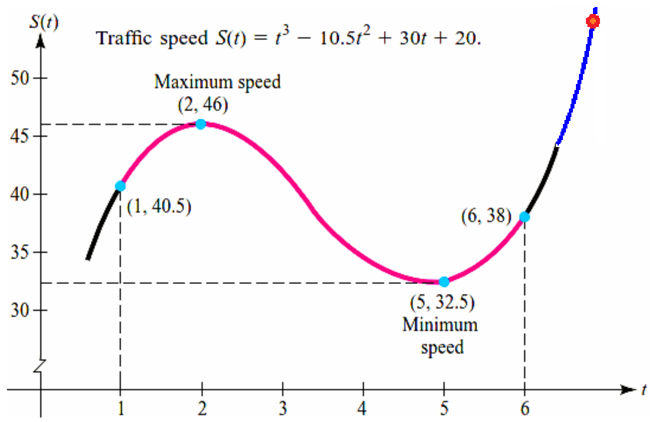
$$S(1) = 40.5 \qquad S(2) = 46 \qquad S(5) = 32.5 \qquad S(6) = 38$$

Since the largest of these values is $S(2) = 46$ and the smallest is $S(5) = 32.5$, we can conclude that the traffic is moving

the fastest at 2:00 pm, when its speed is 46 km/h, and

the slowest at 5:00 pm, when its speed is 32.5 km/h.

The absolute extrema occur at the interior points (which are also the local extrema) instead of at the endpoints. For reference, the graph of S is sketched



Note, if the time domain considered is 1pm-7pm (instead of 1pm-6pm), then the absolute maximum occurs at the end point 7pm, i.e. $S(7) = 58.5$. This value is larger than the local maximum, $S(2) = 46$.

C. Optimization

In this section we give several examples showing applications of calculus to maximum and minimum problems. To solve these examples, go through the following general steps.

In optimization, there is a variable (or quantity) Q to be optimized. Q is a function of some (more than one) variables. The function is called the "**objective function**".

(i). Abstract the problem as a diagram. Sketch and label the various parts. Identify and denote all variables.

(ii). Identify the constraints. Identify Q and formulate the objective function. (Constraints and the objective function will usually come from the geometry or physics of the situation.)

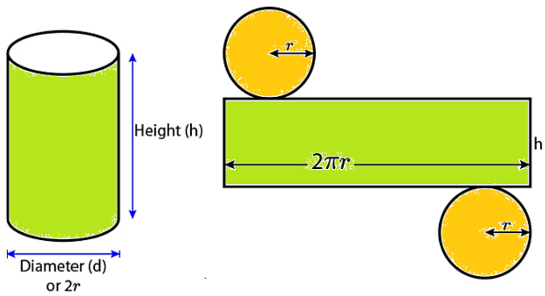
(iii). Use the constraints to perform variable elimination onto the objective function so that Q is reduced to a function of only one variable (called single-variable Q-function). Determine the feasible domain of this single-variable Q-function, i.e. rule out the values for which the problem make no sense.

(iv). Test the critical points, find the desired extremum of the single-variable Q-function by the calculus techniques discussed in the preceding section. Remember, you may have to check for possible endpoint extremum.

(v). Reinterpret your answer in terms of the problem as posed.

Example-7

A company wants to manufacture cylindrical aluminum cans with a volume of 1000 cm^3 (1 liter). What should the radius and height of the can be to minimize the amount of aluminum used?



▶ (i) Volume, $V = \pi r^2 h$

Surface area, $S = 2\pi r h + 2\pi r^2$

▶ (ii) **Constrain**: volume = 1000 cm^3

$V = \pi r^2 h = 1000$

***Q* and the *Q*-function**: minimize the amount of aluminum

Amount of aluminum is $S = 2\pi r h + 2\pi r^2$

▶ (iii) Reduce Q-function to single-variable function (by using constrain)

$S = S(r, h) = 2\pi r h + 2\pi r^2$

Substitute h in to S so that h is eliminated. And hence S reduces to single-variable Q-function.

$S = S(r)$

$$= 2\pi r^2 + 2\pi r \frac{1000}{\pi r^2} = 2\pi r^2 + \frac{2000}{r}$$

• Feasible domain: r should be positive

So, the domain of S is $(0, \infty)$.

> $V = \pi r^2 h = 1000.$
>
> Solve this for h: $h = \dfrac{1000}{\pi r^2}$
>
> *Note*: Solving for r would have involved a square root and a more complicated function.

▶ (iv) Find the critical points for S

Find dS/dr, then solving the equation $dS/dr = 0$ for r.

$$\frac{dS}{dr} = 4\pi r - \frac{2000}{r^2} = 0$$

$$4\pi r^3 = 2000$$

$$r^3 = \frac{500}{\pi}$$

$$r = \left(\frac{500}{\pi}\right)^{1/3} \approx 5.419$$

Substitute this expression into the equation for h to get

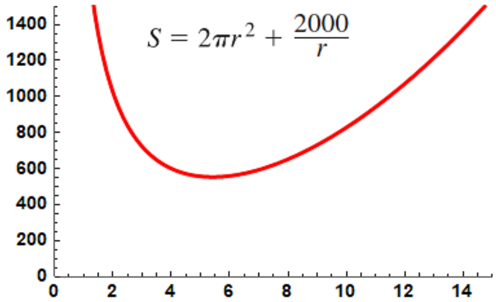
$$h = \frac{1000}{\pi 5.419^2} \approx 10.84$$

Notice that the height of the can is twice its radius.

• To test the critical point
1st derivative test:
Verify that when $r < 5.419$, then $dS/dr < 0$ and when $r > 5.419$, then $dS/dr > 0$.
Hence, there must be a local minimum.

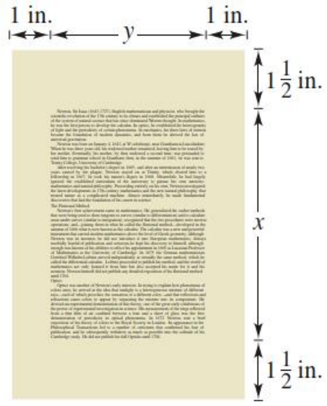
2nd derivative test:
$$\frac{d^2S}{dr^2} = 4\pi + \frac{4000}{r^3}$$
For positive r, the second derivative is always positive, so there is a local minimum

Plotting S using software confirms that there is an absolute minimum at $r = 5.419$ cm.



▶ (v) Reinterpret the answer
To minimize the amount of aluminum used, we should use radius $r = 5.419$ cm and height $h = 10.84$ cm.

Example-8
A rectangular page is to contain 24 square inches of print. The margins at the top and bottom of the page are to be 1.5 inches, and the margins on the left and right are to be 1 inch. What should the dimensions of the page be so that the least amount of paper is used?



Let A be the area to be minimized.
$$A = (x + 3)(y + 2)$$ **Objective function (Q-function)**

The printed area inside the margins is
$$24 = xy.$$ **Constrain**

Solving this equation for y produces $y = 24/x$.
Substitution into the primary equation produces
$$A = (x + 3)\left(\frac{24}{x} + 2\right) = 30 + 2x + \frac{72}{x}$$ **1-variable Q-function**
Because x must be positive, you are interested only in values of A for $x > 0$.

To find the critical numbers, differentiate with respect to x
$$\frac{dA}{dx} = 2 - \frac{72}{x^2}$$
the derivative is zero when $x^2 = 36$, or $x = \pm 6$.
So, the critical numbers are $x = \pm 6$.
You do not have to consider $x = -6$ because it is outside the domain.

First Derivative Test confirms that A is a minimum when $x = 6$.
So, $y = \frac{24}{6} = 4$ and the dimensions of the page should be $x + 3 = 9$ inches by $y + 2 = 6$ inches.

D. L'Hospital's rule and the indeterminate $\frac{0}{0}$ and $\frac{\infty}{\infty}$ forms

1) Consider a limit of the form
$$\lim_{x\to c} \frac{f(x)}{g(x)}$$
where c is either a finite number or $\pm\infty$.
(a) If both $f(x)$ and $g(x)$ approach 0 as x approaches c, such limit is called $\frac{0}{0}$ **indeterminate forms**.
(b) If both $f(x)$ and $g(x)$ increase or decrease without bound as $x \to c$, it is called $\frac{\infty}{\infty}$ **indeterminate forms**.

2) L'Hôpital's Rule

(i) If $\lim_{x\to c} f(x) = 0$ and $\lim_{x\to c} g(x) = 0$, then

$$\lim_{x\to c}\frac{f(x)}{g(x)} = \lim_{x\to c}\frac{f'(x)}{g'(x)}$$

(ii) If $\lim_{x\to c} f(x) = \infty$ and $\lim_{x\to c} g(x) = \infty$, then

$$\lim_{x\to c}\frac{f(x)}{g(x)} = \lim_{x\to c}\frac{f'(x)}{g'(x)}$$

3) When the limit of a quotient f/g leads to $0/0$, ∞/∞ indeterminate forms, then take derivatives of the numerator and the denominator and try again. Continue to differentiate f and g as long as still getting the indeterminate forms. But as soon as one or the other of these derivatives is different from 0 or ∞ at $x = c$, we stop differentiating. L'Hôpital's Rule does not apply when either the numerator or denominator has a finite nonzero limit.

Example-9

(a) $\lim_{x\to 0}\frac{x-\sin x}{x^3}$ $\quad\frac{0}{0}$

$= \lim_{x\to 0}\frac{1-\cos x}{3x^2}$ $\quad$ Still $\frac{0}{0}$

$= \lim_{x\to 0}\frac{\sin x}{6x}$ $\quad$ Still $\frac{0}{0}$

$= \lim_{x\to 0}\frac{\cos x}{6} = \frac{1}{6}$ $\quad$ Not $\frac{0}{0}$; limit is found.

(b) $\lim_{x\to 0}\frac{1-\cos x}{x+x^2}$ $\quad\frac{0}{0}$

$= \lim_{x\to 0}\frac{\sin x}{1+2x} = \frac{0}{1} = 0$ $\quad$ Not $\frac{0}{0}$; limit is found.

Up to now the calculation is correct, but if we continue to differentiate in an attempt to apply l'Hôpital's Rule once more, we get

$$\lim_{x\to 0}\frac{1-\cos x}{x+x^2} = \lim_{x\to 0}\frac{\sin x}{1+2x} = \lim_{x\to 0}\frac{\cos x}{2} = \frac{1}{2},$$

which is wrong.

L'Hôpital's Rule can only be applied to limits which give indeterminate forms, and $0/1$ is not an indeterminate form.

(c) $\lim_{x\to\infty}\frac{x-2x^2}{3x^2+5x} = \lim_{x\to\infty}\frac{1-4x}{6x+5}$

$= \lim_{x\to\infty}\frac{-4}{6} = -\frac{2}{3}$

Example-10

Sometimes limits of other forms can be handled by using algebra to convert them to $0/0$, ∞/∞ indeterminate forms.

(a) $\lim_{x\to\infty}\left(x\sin\frac{1}{x}\right)$ $\quad\infty\cdot 0$ $\quad$ Let $h = 1/x$

$= \lim_{h\to 0^+}\left(\frac{1}{h}\sin h\right)$ $\quad 0/0$

$= \lim_{h\to 0^+}\left(\frac{\cos h}{1}\right)$

$= 1$

(b) $\lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{1}{x}\right)$ $\quad\infty-\infty$

$= \lim_{x\to 0}\frac{x-\sin x}{x\sin x}$ $\quad\frac{0}{0}$

$= \lim_{x\to 0}\frac{1-\cos x}{\sin x + x\cos x}$ $\quad$ Still $\frac{0}{0}$

$= \lim_{x\to 0}\frac{\sin x}{2\cos x - x\sin x} = \frac{0}{2} = 0$

E. Newton's method in roots finding

1) **Isaac Newton** first described the method for approximating the real zeros of a function in his text Method of Fluxions. Although the book was written in 1671, it was not published until 1736. Meanwhile, in 1690, **Joseph Raphson** (1648–1715) published a paper describing a method for approximating the real zeros of a function that was very similar to Newton's. For this reason, the method is often referred to as the **Newton-Raphson method**.

2) Let $f(c) = 0$ where f is differentiable on an open interval containing c. Then, to approximate c, use these steps.

i) Make an initial estimate x_1 that is close to c. (A graph is helpful.)

ii) Determine a new approximation

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

iii) When $|x_n - x_{n+1}|$ is within the desired accuracy, let x_{n+1} serve as the final approximation. Otherwise, return to Step ii) and calculate a new approximation.

3) A theorem from advanced calculus says that if

$$\left|\frac{f(x)f''(x)}{(f'(x))^2}\right| < 1$$

for all x in an interval about a root r, then the method will converge to r for any starting value x_0 in that interval. Note that this condition is satisfied if the graph of f is not too horizontal near where it crosses the x-axis.

$$f'(x_1) \approx \frac{f(x_2)-f(x_1)}{x_2-x_1}$$

For $f(x_2) = 0,\ x_2 - x_1 \approx -\frac{f(x_1)}{f'(x_1)} \Rightarrow x_2 \approx x_1 - \frac{f(x_1)}{f'(x_1)}$

Example-11

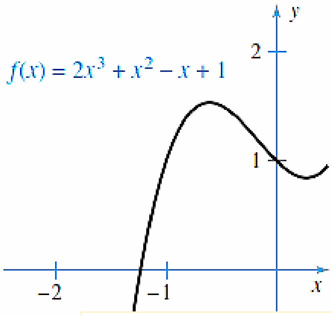
Use Newton's Method to approximate the zeros of

$$f(x) = 2x^3 + x^2 - x + 1.$$

Continue the iterations until two successive approximations differ by less than 0.0001.

Use $x_1 = -1.2$ as the initial guess.

From the graph, you can observe that the function has only one zero, which occurs near $x = -1.2$.



Next, differentiate f and form the iterative formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{2x_n^3 + x_n^2 - x_n + 1}{6x_n^2 + 2x_n - 1}.$$

The calculations are shown in the table.

n	x_n	$f(x_n)$	$f'(x_n)$	$\frac{f(x_n)}{f'(x_n)}$	$x_n - \frac{f(x_n)}{f'(x_n)}$	$\lvert x_n - x_{n-1}\rvert$
1	-1.20000	0.18400	5.24000	0.03511	-1.23511	n.a.
2	-1.23511	-0.00771	5.68276	-0.00136	-1.23375	0.03511
3	-1.23375	0.00001	5.66533	0.00000	-1.23375	0.00136
4	-1.23375					0.00000

Because two successive approximations differ by less than the required 0.0001, you can estimate the zero of f to be -1.23375.

Example-12

Consider the equation

$$x^{1/3} = 0$$

which has $x = 0$ as its only solution. Try to approximate this solution by Newton's Method with a starting value of $x_1 = 0.1$.

The function $f(x) = x^{1/3}$ is not differentiable at $x = 0$.

$f'(x) = \frac{1}{3}x^{-2/3}$, the iterative formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n{}^{1/3}}{\frac{1}{3}x_n{}^{-2/3}} = x_n - 3x_n = -2x_n$$

The calculations are shown in the table. This table indicate that x_n continues to increase in magnitude as $n \to \infty$, and so the limit of the sequence does not exist.

n	x_n	$f(x_n)$	$f'(x_n)$	$\frac{f(x_n)}{f'(x_n)}$	$x_n - \frac{f(x_n)}{f'(x_n)}$
1	0.10000	0.46416	1.54720	0.30000	−0.20000
2	−0.20000	−0.58480	0.97467	−0.60000	0.40000
3	0.40000	0.73681	0.61401	1.20000	−0.80000
4	−0.80000	−0.92832	0.3680	−2.40000	1.60000

Notice,

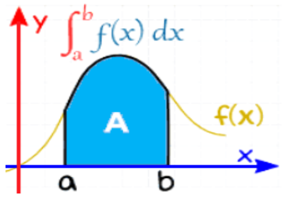
$$f(x) = x^{1/3}, \quad f'(x) = \frac{1}{3}x^{-2/3}, \quad f''(x) = -\frac{2}{9}x^{-5/3}$$

and

$$\left|\frac{f(x)\,f''(x)}{[f'(x)]^2}\right| = \left|\frac{x^{1/3}(-2/9)(x^{-5/3})}{(1/9)(x^{-4/3})}\right| = 2$$

which is not less than 1 for any value of x, so you cannot conclude that Newton's Method will converge.

Integrals



A. Antidifferentiation, antiderivatives, indefinite integrals

1) Antidifferentiation:

The derivative of x^2 with respect to x is $2x$ i.e. $\frac{d}{dx}x^2 = 2x$. Conversely, given that an unknown expression has derivative $2x$, it is clear that the unknown expression could be x^2 i.e. $\frac{d}{dx}F? = 2x \to F? = x^2$. The process of finding a function from its derivative is called **antidifferentiation**.

2) Antiderivatives, general antiderivative:

Now consider the functions $f(x) = x^2 + 1$ and $g(x) = x^2 - 7$. We have $f'(x) = 2x$ and $g'(x) = 2x$. So the two different functions have the same derivative function. Both $x^2 + 1$ and $x^2 - 7$ are said to be **antiderivatives** of $2x$. The **general antiderivative** of $2x$ is $x^2 + C$, where C is an arbitrary real number.

3) Indefinite integral:

We use the notation of Leibniz to state this with symbols:

$$\int 2x\,dx = x^2 + C$$

This is read as 'the general antiderivative of $2x$ with respect to x is equal to $x^2 + C$' or as 'the **indefinite integral** of $2x$ with respect to x is $x^2 + C$'. It should be clearly understood that there is not a unique antiderivative for a given function.

4) Integrand, antiderivative, constant of integration:

In general, expressing this symbolically,

$$\int f(x)\,dx = F(x) + C \text{ if and only if } \frac{dF}{dx} = f(x).$$

Here,

- C, called the **constant of integration**, is an arbitrary real number;
- the function $f(x)$ is called the **integrand**;
- the function $F(x)$ is called the (general) **antiderivative** of $f(x)$.

5) Integration is the "inverse" of differentiation, likewise differentiation is the "inverse" of integration.

$$\int F'(x)\,dx = F(x) + C.$$

Moreover, if $\int f(x)\,dx = F(x) + C$, then

$$\frac{d}{dx}\left[\int f(x)\,dx\right] = f(x).$$

Integrals of Elementary Functions

The following results can be demonstrated by differentiating both sides to produce an identity. In each case, an arbitrary constant c (which has been omitted here) should be added.

1. $\int u^n \, du = \dfrac{u^{n+1}}{n+1} \quad n \neq -1$
2. $\int \dfrac{du}{u} = \ln |u|$
3. $\int \sin u \, du = -\cos u$
4. $\int \cos u \, du = \sin u$
5. $\int \tan u \, du = \ln |\sec u| = -\ln |\cos u|$
6. $\int \cot u \, du = \ln |\sin u|$
7. $\int \sec u \, du = \ln |\sec u + \tan u| = \ln |\tan(u/2 + \pi/4)|$
8. $\int \csc u \, du = \ln |\csc u - \cot u| = \ln |\tan u/2|$
9. $\int \sec^2 u \, du = \tan u$
10. $\int \csc^2 u \, du = -\cot u$
11. $\int \sec u \tan u \, du = \sec u$
12. $\int \csc u \cot u \, du = -\csc u$
13. $\int a^u \, du = \dfrac{a^u}{\ln a} \quad a > 0, a \neq 1$
14. $\int e^u \, du = e^u$
15. $\int \dfrac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \dfrac{u}{a}$ or $-\cos^{-1} \dfrac{u}{a}$
16. $\int \dfrac{du}{\sqrt{u^2 \pm a^2}} = \ln \left| u + \sqrt{u^2 \pm a^2} \right|$
17. $\int \dfrac{du}{u^2 + a^2} = \dfrac{1}{a} \tan^{-1} \dfrac{u}{a}$ or $-\dfrac{1}{a} \cot^{-1} \dfrac{u}{a}$
18. $\int \dfrac{du}{u^2 - a^2} = \dfrac{1}{2a} \ln \left| \dfrac{u-a}{u+a} \right|$
19. $\int \sqrt{u^2 \pm a^2} \, du = \dfrac{u}{2} \sqrt{u^2 \pm a^2} \pm \dfrac{a^2}{2} \ln \left| u + \sqrt{u^2 \pm a^2} \right|$
20. $\int \sqrt{a^2 - u^2} \, du = \dfrac{u}{2} \sqrt{a^2 - u^2} + \dfrac{a^2}{2} \sin^{-1} \dfrac{u}{a}$

In particular for linear function $(ax + b)$:

$f(x)$	$\int f(x)\,dx$	
$(ax+b)^n$	$\dfrac{1}{a(n+1)}(ax+b)^{n+1} + c$	where $n \neq -1$
$\dfrac{1}{ax+b}$	$\dfrac{1}{a} \ln(ax+b) + c$	for $ax + b > 0$
e^{ax+b}	$\dfrac{1}{a} e^{ax+b} + c$	

$f(x)$	$\int f(x)\,dx$
$\sin(ax+b)$	$-\dfrac{1}{a}\cos(ax+b) + c$
$\cos(ax+b)$	$\dfrac{1}{a}\sin(ax+b) + c$

Laws for Antiderivatives

1. $\int 0 \, dx = C$

2. $\int 1 \, dx = x + C$

3. $\int a \, dx = ax + C$

4. $\int af(x) \, dx = a \int f(x) \, dx$

5. $\int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$

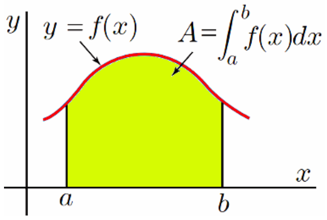
6. Quick Formula : $\int \dfrac{g'(x)}{g(x)} dx = \ln |g(x)| + C$

7. Substitution method: $\int f(g(x)) g'(x) \, dx = \int f(u) \, du$

B. Area, Riemann sum, Definite integrals

1) Area under a graph

If $f(x)$ is a continuous nonnegative function on the interval $a \le x \le b$, we refer to the area of the region shown in the figure as **the area under the graph of $f(x)$ from a to b.**



2) Theorem I: area = definite integral

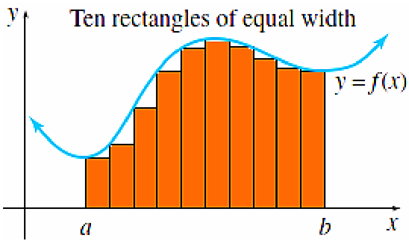
If $f(x)$ is a continuous nonnegative function on the interval $a \leqslant x \leqslant b$, then the area under the graph of $f(x)$ from a to b *is* equal to **the definite integral of f from a to b**:

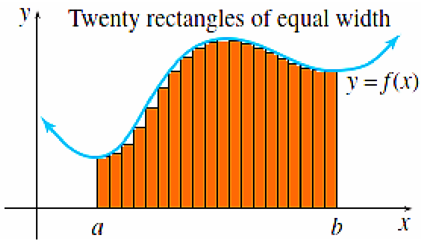
$$\begin{bmatrix}\text{Area under the graph of } f \\ \text{from } x = a \text{ to } x = b\end{bmatrix} = \int_a^b f(x)\,dx = F(b) - F(a)$$

where F is an antiderivative of f.

3) Approximate area using many rectangles

We can estimate the area under a graph to any desired degree of accuracy by using rectangles whose total area is approximately equal to the area we want to computed. The thinner the rectangles are, the smaller the mismatch between the rectangles and the area under the graph is.





4) Riemann sums

Let f be defined on the interval $[a, b]$, and let Δ be a partition of $[a, b]$ given by

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

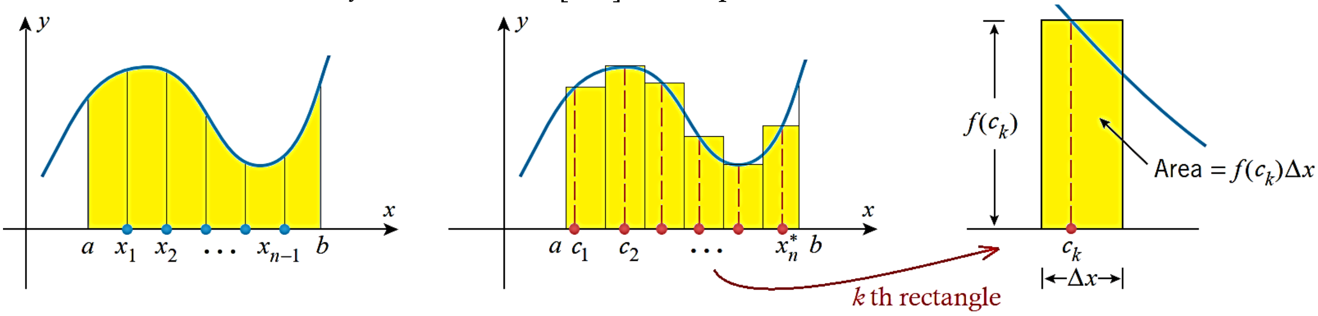
where Δx_i is the width of the i-th subinterval $[x_{i-1}, x_i]$.

If c_i is any point in the i-th subinterval, then the sum

$$R_n = \sum_{i=1}^{n} f(c_i)\Delta x_i, \quad x_{i-1} \le c_i \le x_i$$

is called a **Riemann sum** of f on the interval $[a, b]$ for the partition Δ.

Partition $a \leqslant x \leqslant b$ into n subintervals of width $\Delta x = \dfrac{b-a}{n}$ each.



There are many such sums, depending on the partition we choose, and the choices of the points c_i's in the subintervals (e.g. it can be the ***left endpoint***, ***right endpoint*** or ***midpoint*** of each subinterval).

5) Theorem II: definite integral = limit of Riemann sums

Let f be a continuous function on the interval $[a, b]$, with an antiderivative F. Then, the Riemann sums R_n approach the definite integral of f on $[a, b]$, as the number of subintervals increases indefinitely. That is,

$$\lim_{\Delta x \to 0} \sum_{i=1}^{n} f(c_i)\Delta x_i = \int_a^b f(x)\, dx = F(b) - F(a)$$

Note: $\Delta x = \frac{b-a}{n}$, so $\lim\limits_{\Delta x \to 0}$ implies $\lim\limits_{n \to \infty}$ i.e. as the width of the partition approaches 0, the number of subintervals in a partition approaches infinity.

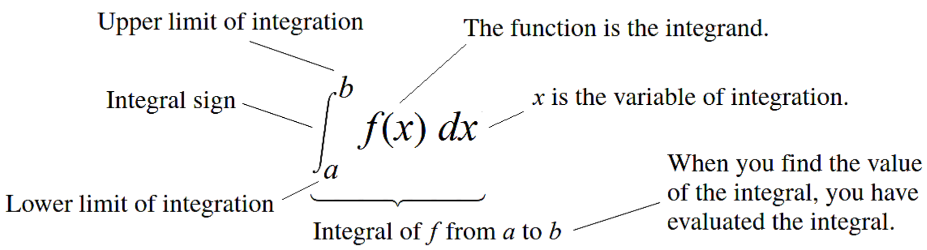
Note: combining Theorem I & II: area = definite integral = limit of Riemann sums

C. The Fundamental Theorem of Calculus, Evaluating definite integrals

1) The **definite integral** of f over $[a, b]$ is

$$\int_a^b f(x)\, dx$$

which is read as "the integral from a to b of f of x dee x" or sometimes as "the integral from a to b of f of x with respect to x." The component parts in the integral symbol also have names:



2) If f is continuous on $[a, b]$, and $x \in (a, b)$ then

(i) $\int_a^b f(x)\, dx = F(b) - F(a)$ where $F(x)$ is any antiderivarive of $f(x)$, that is $F' = f$.

(ii) If $F(x) = \int_a^x f(t)\, dt$, then $F'(x) = f(x)$, or $\frac{d}{dx}\int_a^x f(t)\, dt = f(x)$.

This is known as **The Fundamental Theorem of Calculus**.

3) Evaluating definite integral by computing the limit of the Riemann sums would be very complicated. Fortunately, there is an easier way of performing this computation without having to use the limit of a sum, thanks to the deep result of FTC (i) above connecting the definite integral to antidifferentiation. [see E.g. 7]

4) In evaluating definite integrals using The Fundamental Theorem of Calculus, after obtaining an antiderivative of f, it is not necessary to include a constant of integration in the antiderivative since the C's cancel.

$$\int_a^b f(x)\, dx = \Big[F(x) + C\Big]_a^b = [F(b) + C] - [F(a) + C] = F(b) - F(a)$$

D. Some properties of definite integrals

1. $\int_a^b \{f(x) \pm g(x)\}dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$
2. $\int_a^b Af(x)dx = A\int_a^b f(x)dx$ where A is any constant
3. $\int_a^b f(x)\,dx = \int_a^k f(x)\,dx + \int_k^b f(x)\,dx$ provided $f(x)$ is integrable in $[a, k]$ and $[k, b]$
4. $\int_a^b f(x)dx = -\int_b^a f(x)dx$
5. $\int_a^a f(x)dx = 0$

Example-1

Find an antiderivative of each of the following:

a $\sin\left(3x - \frac{\pi}{4}\right)$ **b** e^{3x+4} **c** $6x^3 - \frac{2}{x^2}$

a $\sin\left(3x - \frac{\pi}{4}\right)$ is of the form $\sin(ax + b)$

$$\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + c$$

$$\therefore \int \sin\left(3x - \frac{\pi}{4}\right)dx = -\frac{1}{3}\cos\left(3x - \frac{\pi}{4}\right) + c$$

b e^{3x+4} is of the form e^{ax+b}

$$\int e^{ax+b}\,dx = \frac{1}{a}e^{ax+b} + c$$

$$\therefore \int e^{3x+4}\,dx = \frac{1}{3}e^{3x+4} + c$$

c $\int 6x^3 - \frac{2}{x^2}\,dx = \int 6x^3 - 2x^{-2}\,dx$

$$= \frac{6x^4}{4} + 2x^{-1} + c$$

$$= \frac{3}{2}x^4 + \frac{2}{x} + c$$

Example-2

Evaluate each of the following integrals:

a $\int_0^{\frac{\pi}{8}} \sec^2(2x)\,dx$ **b** $\int_0^1 \sqrt{2x+1}\,dx$ **c** $\int_{-2}^{-1} \frac{1}{4x+2}\,dx.$

a $\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + c$

$$\therefore \int_0^{\frac{\pi}{8}} \sec^2(2x)\,dx = \left[\frac{1}{2}\tan(2x)\right]_0^{\frac{\pi}{8}}$$

$$= \frac{1}{2}\left(\tan\left(\frac{\pi}{4}\right) - \tan 0\right)$$

$$= \frac{1}{2}(1-0) = \frac{1}{2}$$

b $\int_0^1 \sqrt{2x+1}\,dx$

$$= \int_0^1 (2x+1)^{\frac{1}{2}}\,dx$$

$$= \left[\frac{1}{2 \times \frac{3}{2}}(2x+1)^{\frac{3}{2}}\right]_0^1$$

$$= \frac{1}{3}\left((2+1)^{\frac{3}{2}} - 1^{\frac{3}{2}}\right)$$

$$= \frac{1}{3}\left(3^{\frac{3}{2}} - 1\right) = \frac{1}{3}(3\sqrt{3} - 1)$$

c $\int_{-2}^{-1} \frac{1}{4x+2}\,dx = \left[\frac{1}{4}\log_e|4x+2|\right]_{-2}^{-1}$

$$= \frac{1}{4}(\log_e|-2| - \log_e|-6|)$$

$$= \frac{1}{4}\log_e\left(\frac{1}{3}\right)$$

$$= -\frac{1}{4}\log_e 3$$

Example-3

Evaluate (a) $\int \frac{1}{\sqrt{18-9x^2}}dx$ (b) $\int_0^{\pi} \sin^2 x\,dx$

(a) $\int \frac{1}{\sqrt{9(2-x^2)}}dx = \frac{1}{3}\int \frac{1}{\sqrt{(\sqrt{2})^2 - x^2}}dx = \frac{1}{3}\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$

(b) Recall the identity

$$\sin^2 x = \tfrac{1}{2}(1 - \cos 2x) \quad \text{and} \quad \cos^2 x = \tfrac{1}{2}(1 + \cos 2x)$$

$$\int_0^{\pi} \sin^2 x \, dx = \tfrac{1}{2}\int_0^{\pi} (1 - \cos 2x)\, dx = \left[\tfrac{1}{2}\left(x - \tfrac{1}{2}\sin 2x\right)\right]_0^{\pi}$$

$$= \tfrac{1}{2}\left(\pi - \tfrac{1}{2}\sin 2\pi\right) - \tfrac{1}{2}\left(0 - \tfrac{1}{2}\sin 0\right) = \tfrac{1}{2}\pi$$

Example-4

(a) $$\int \frac{(1+x)^2}{\sqrt{x}}\, dx = \int \frac{1+2x+x^2}{x^{1/2}}\, dx = \int (x^{-1/2} + 2x^{1/2} + x^{3/2})\, dx$$

$$= 2x^{1/2} + \frac{4}{3}x^{3/2} + \frac{2}{5}x^{5/2} + C$$

(b) $$\int \frac{x^2+2x}{(x+1)^2}\, dx = \int \left[1 - \frac{1}{(x+1)^2}\right] dx = x + \frac{1}{x+1} + C' \quad \text{(may stop here)}$$

(cemplete square)

$$= \frac{x^2}{x+1} + 1 + C' = \frac{x^2}{x+1} + C$$

Example-5 (Quick)

(a) $$\int \frac{2x}{x^2+1} dx = \ln|x^2+1| + C = \ln(x^2+1) + C$$

The absolute value sign was dropped because $x^2 + 1 \geq 0$.

(b) $$\int \frac{x^2}{x^3+5} dx = \frac{1}{3}\int \frac{3x^2}{x^3+5} dx = \frac{1}{3}\ln|x^3+5| + C$$

Example-6

(a) Find the area of the shaded region under the graph of $f(x) = 3x^2 + e^x$, from $x = -1$ to $x = 1$.

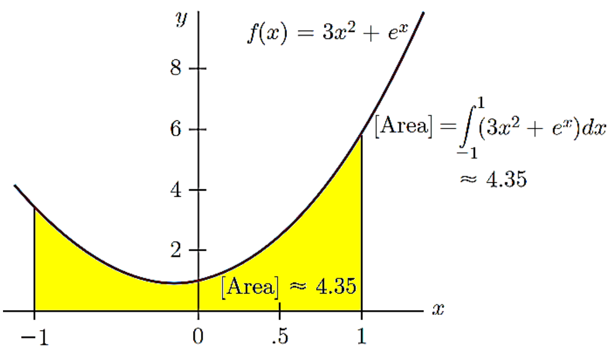
(a)

$$\int_{-1}^{1} (3x^2 + e^x)dx = (x^3 + e^x)\Big|_{-1}^{1}$$

$$= (1 + e^1) - (-1 + e^{-1})$$

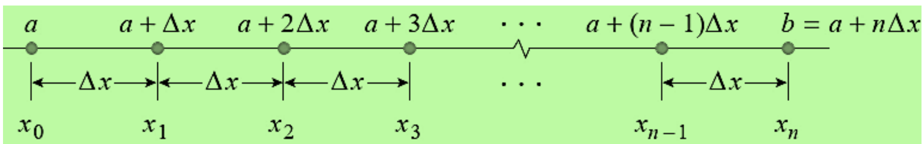
$$= 2 + e - 1/e \approx 4.35.$$

Thus, the area is approximately 4.35.



Example-7 (optional)

From the Riemann sums, choose the right endpoint of each subinterval to find the area under the graph of $f(x) = x^2$ in the interval $[0, 1]$.



THEOREM

(a) $$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

(b) $$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

(c) $$\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$$

The length of each subinterval is

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$$

Choosing the right endpoint,

$$c_k = a + k\Delta x = \frac{k}{n}$$

Thus,

$$\sum_{k=1}^{n} f(c_k)\Delta x = \sum_{k=1}^{n} (c_k)^2 \Delta x = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3}\sum_{k=1}^{n} k^2$$

$$= \frac{1}{n^3}\left[\frac{n(n+1)(2n+1)}{6}\right] = \frac{1}{6}\left(\frac{n}{n}\cdot\frac{n+1}{n}\cdot\frac{2n+1}{n}\right)$$

$$= \frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)$$

Area $= \int_0^1 x^2\,dx$

$$= \lim_{n\to+\infty}\sum_{k=1}^{n} f(c_k)\Delta x = \lim_{n\to+\infty}\left[\frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right] = \frac{1}{3}$$

If the integral interval is $[0, x]$,

$\Delta x = \frac{x-0}{n} = \frac{x}{n}$; $c_k = \frac{kx}{n}$

Riemann sums,

$$\sum_{k=1}^{n}\left(\frac{kx}{n}\right)^2\frac{x}{n} = \sum_{k=1}^{n}\frac{k^2x^3}{n^3}$$

$$= \frac{x^3}{n^3}\sum_{k=1}^{n} k^2 = \frac{x^3}{n^3}\frac{n(n+1)(2n+1)}{6}$$

$$= \frac{x^3}{n^2}\frac{(n+1)(2n+1)}{6}$$

$$= \frac{x^3}{6}\frac{2n^2+3n+1}{n^2} = \frac{x^3}{6}\left(2+\frac{3}{n}+\frac{1}{n^2}\right)$$

As $n\to\infty$, the Riemann sums approaches $\frac{x^3}{3}$.

Hence, $\int_0^x x^2\,dx = \frac{x^3}{3}$.

Example-8 (Fundamental Theorem of Calculus)

(a) Find the derivative of $g(x) = \int_1^x \frac{dt}{t^3+2}$.

(b) Let $F(x) = \int_1^{\sqrt{x}} \sin t\,dt$, find $F'(x)$.

(a) According to the Fundamental Theorem of Calculus: $g'(x) = \frac{1}{x^3+2}$

(b) Letting $u(x) = \sqrt{x}$, we have $F(x) = \int_1^{u(x)} \sin t dt$.

$$F'(x) = \sin(u(x))\frac{du}{dx} = \sin(u(x))\cdot\left(\frac{1}{2}x^{-1/2}\right) = \frac{\sin\sqrt{x}}{2\sqrt{x}}.$$

E. Integration techniques: substitution, by parts, partial fractions

(i) By substitution

Example-9

Evaluate $\int x\, e^{x^2}\,dx$

Let $u = x^2 \implies du = 2x\,dx \implies x\,dx = \frac{1}{2}du$

$$\int x\,e^{x^2}\,dx = \frac{1}{2}\int e^u du = \frac{1}{2}e^u = \frac{1}{2}e^{x^2} + c$$

Example-10

(a) Find $\int x\sin(x^2)dx$.

Let $u = x^2$. Then $du = 2x\,dx$. So, $x\,dx = \frac{1}{2}du$. By substitution,

$$\int x\sin(x^2)dx = \int \sin u\ (\tfrac{1}{2})du = \tfrac{1}{2}(-\cos u) + C = -\tfrac{1}{2}\cos(x^2) + C$$

(b) $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx.$

Let $u=\sqrt{x}=x^{1/2}$. Then $du=\frac{1}{2}x^{-1/2}dx$. So, $2\,du=\frac{1}{\sqrt{x}}dx$. Thus,

$$\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx = 2\int \cos u\,du = 2\sin u + C = 2\sin(\sqrt{x})+C$$

(c) $\int x\sec^2(4x^2-5)\,dx$.

Let $u=4x^2-5$. Then $du=8x\,dx$, $\frac{1}{8}du = x\,dx$. Thus,

$$\int x\sec^2(4x^2-5)\,dx = \frac{1}{8}\int \sec^2 u\,du = \frac{1}{8}\tan u + C = \frac{1}{8}\tan(4x^2-5)+C$$

(d) Evaluate $\int \cos^3 x\ dx$

Simply substituting $u=\cos x$ isn't helpful, since then $du=-\sin x\,dx$.

Use the identity $\sin^2 x + \cos^2 x = 1$:

$$\cos^3 x = \cos^2 x \cdot \cos x = (1-\sin^2 x)\cos x$$

Substituting $u=\sin x$, so $du=\cos x\,dx$:

$$\int \cos^3 x\,dx = \int \cos^2 x\cdot\cos x\,dx = \int (1-\sin^2 x)\cos x\,dx$$
$$= \int (1-u^2)\,du = u - \tfrac{1}{3}u^3 + C = \sin x - \tfrac{1}{3}\sin^3 x + C$$

(ii) By parts (the table-method or fast-method)

The formula for integration by parts is commonly given as

$$\int u\,dv = uv - \int v\,du$$

Integrals of the form $\int f(x)g(x)\,dx$, in which f can be differentiated repeatedly to become zero and g can be integrated repeatedly without difficulty, are natural candidates for integration by parts. However, if many repetitions are required, the calculations can be cumbersome. In situations like this, there is a way to organize the calculations that saves a great deal of work. It is called tabular integration. We will only learn this method here.

Example-11

Evaluate (a) $\int x^2 e^x\,dx$ (b) $\int x^3 \sin x\,dx$

(a) With $f(x)=x^2$ and $g(x)=e^x$, we list:

$f(x)$ and its derivatives		**$g(x)$ and its integrals**
x^2	$(+)$	e^x
$2x$	$(-)$	e^x
2	$(+)$	e^x
0		e^x

We combine the products of the functions connected by the arrows according to the operation signs above the arrows to obtain

$$\int x^2 e^x\,dx = x^2e^x - 2xe^x + 2e^x + C.$$

(b)

$f(x)$ and its derivatives		$g(x)$ and its integrals
x^3	$(+)$	$\sin x$
$3x^2$	$(-)$	$-\cos x$
$6x$	$(+)$	$-\sin x$
6	$(-)$	$\cos x$
0		$\sin x$

$$\int x^3 \sin x\, dx = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C.$$

The horizontal arrow at the end

Each diagonal arrow indicates a multiplication between the terms at either end, the result of which is then added up, i.e. sum product. The final horizontal arrow indicates that the two terms at either end are to be multiplied together and then integrated before adding to the "sum product" from the previous levels.

Example-12

Evaluate (a) $\int \ln x\, dx$ (b) $\int x^2 \ln 4x\, dx$

(a)

$\ln x$	$+$	1
$\frac{1}{x}$	$-$	x

$$\int \ln x\, dx = x \ln x - \int 1\, dx = x \ln x - x + C$$

(b)

$\ln 4x$	$+$	x^2
$\frac{1}{x}$	$-$	$\frac{x^3}{3}$

$$\begin{aligned}\int x^2 \ln 4x\, dx &= \frac{x^3}{3}\ln 4x - \frac{1}{3}\int x^2\, dx \\ &= \frac{x^3}{3}\ln 4x - \frac{1}{9}x^3 + C \\ &= x^3\left(\frac{1}{3}\ln 4x - \frac{1}{9}\right) + C\end{aligned}$$

Example-13 (integrand reappear)

Evaluate $\int e^{2x} \sin 3x\, dx$.

e^{2x}	$+$	$\sin 3x$
$2e^{2x}$	$-$	$-\frac{1}{3}\cos 3x$
$4e^{2x}$	$+$	$-\frac{1}{9}\sin 3x$

$$\begin{aligned}\mathcal{J} &= \int e^{2x} \sin 3x\, dx \\ &= -\frac{1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x - \frac{4}{9}\int e^{2x}\sin 3x\, dx \\ \mathcal{J} &= -\frac{1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x - \frac{4}{9}\mathcal{J} \\ \frac{13}{9}\mathcal{J} &= -\frac{1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x \\ \mathcal{J} &= -\frac{9}{39}e^{2x}\cos 3x + \frac{2}{13}e^{2x}\sin 3x + C \\ &= \frac{1}{13}e^{2x}(-3\cos 3x + 2\sin 3x) + C\end{aligned}$$

(iii) By partial fractions

If $g(x)$ and $h(x)$ are polynomials, then

$f(x) = \dfrac{g(x)}{h(x)}$ is a **rational function**;

e.g. $\quad f(x) = \dfrac{4x+2}{x^2-1}$

- If the degree of $g(x)$ is less than the degree of $h(x)$, then $f(x)$ is a **proper fraction**.
- If the degree of $g(x)$ is greater than or equal to the degree of $h(x)$, then $f(x)$ is an **improper fraction**.

A rational function may be expressed as a sum of simpler functions by resolving it into what are called **partial fractions**. For example:

$$\frac{4x+2}{x^2-1} = \frac{3}{x-1} + \frac{1}{x+1}$$

We will see that this is a useful technique for integration.

Proper fractions

We only consider examples where the denominators have factors that are either degree 1 (linear) or degree 2 (quadratic).

Method of Partial Fractions ($f(x)/g(x)$ Proper)

1. Let $x - r$ be a linear factor of $g(x)$. Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. Then, to this factor, assign the sum of the m partial fractions:

$$\frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \cdots + \frac{A_m}{(x-r)^m}.$$

Do this for each distinct linear factor of $g(x)$.

2. Let $x^2 + px + q$ be a quadratic factor of $g(x)$. Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:

$$\frac{B_1x + C_1}{x^2+px+q} + \frac{B_2x + C_2}{(x^2+px+q)^2} + \cdots + \frac{B_nx + C_n}{(x^2+px+q)^n}.$$

Do this for each distinct quadratic factor of $g(x)$ that cannot be factored into linear factors with real coefficients.

Examples of resolving a proper fraction into partial fractions:

(i) Distinct linear factors

$$\frac{3x-4}{(2x-3)(x+5)} = \frac{A}{2x-3} + \frac{B}{x+5}$$

(ii) Repeated linear factor

$$\frac{3x-4}{(2x-3)(x+5)^2} = \frac{A}{2x-3} + \frac{B}{x+5} + \frac{C}{(x+5)^2}$$

(iii) Irreducible quadratic factor

$$\frac{3x-4}{(2x-3)(x^2+5)} = \frac{A}{2x-3} + \frac{Bx+C}{x^2+5}$$

(iv) Repeated quadratic factor

$$\frac{3x-4}{(2x-3)(x^2+5)^2} = \frac{A}{2x-3} + \frac{Bx+C}{x^2+5} + \frac{Dx+E}{(x^2+5)^2}$$

To resolve an algebraic fraction into its partial fractions:

Step-1 Write a statement of identity between the original fraction and a sum of the appropriate number of partial fractions.

Step-2 Express the sum of the partial fractions as a single fraction, and note that the numerators of both sides are equivalent.

Step-3 Find the values of the introduced constants $A, B, C, \ldots$ by substituting appropriate values for x or by equating coefficients.

Example-14

Resolve $\dfrac{2x+10}{(x+1)(x-1)^2}$ into partial fractions.

$$\frac{2x+10}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

This gives the equation

$$2x + 10 = A(x-1)^2 + B(x+1)(x-1) + C(x+1)$$

Let $x = 1$: $\quad 12 = 2C \quad \therefore \quad C = 6$

Let $x = -1$: $\quad 8 = 4A \quad \therefore \quad A = 2$

Let $x = 0$: $\quad 10 = A - B + C$

$\therefore \quad B = A + C - 10 = -2$

Hence $\dfrac{2x+10}{(x+1)(x-1)^2} = \dfrac{2}{x+1} - \dfrac{2}{x-1} + \dfrac{6}{(x-1)^2}$.

Example-15

Resolve $\dfrac{x^2+6x+5}{(x-2)(x^2+x+1)}$ into partial fractions.

$$\frac{x^2+6x+5}{(x-2)(x^2+x+1)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+x+1}$$

This gives the equation

$$x^2 + 6x + 5 = A(x^2+x+1) + (Bx+C)(x-2) \qquad \textbf{(1)}$$

Subsituting $x = 2$:

$$2^2 + 6(2) + 5 = A(2^2 + 2 + 1)$$

$$21 = 7A \quad \therefore \quad A = 3$$

We can rewrite equation (1) as

$$\begin{aligned} x^2 + 6x + 5 &= A(x^2+x+1) + (Bx+C)(x-2) \\ &= A(x^2+x+1) + Bx^2 - 2Bx + Cx - 2C \\ &= (A+B)x^2 + (A-2B+C)x + A - 2C \end{aligned}$$

Since $A = 3$, this gives

$$x^2 + 6x + 5 = (3+B)x^2 + (3-2B+C)x + 3 - 2C$$

Equate coefficients:

$3 + B = 1 \ \therefore B = -2 \quad$ and $\quad 3 - 2C = 5 \ \therefore C = -1$

Check: $3 - 2B + C = 3 - 2(-2) + (-1) = 6$

Therefore

$$\frac{x^2+6x+5}{(x-2)(x^2+x+1)} = \frac{3}{x-2} + \frac{-2x-1}{x^2+x+1} = \frac{3}{x-2} - \frac{2x+1}{x^2+x+1}$$

Note: The values of B and C could also be found by substituting $x = 0$ and $x = 1$ in equation (1).

Improper fractions

Improper algebraic fractions can be expressed as a sum of partial fractions by first dividing the denominator into the numerator to produce a quotient and a proper fraction. This proper fraction can then be resolved into its partial fractions using the techniques just introduced.

Example-16

Express $\dfrac{x^5+2}{x^2-1}$ as partial fractions.

Dividing through:

$$\begin{array}{r|l} & x^3 + x \\ \hline x^2-1 & x^5 + 2 \\ & x^5 - x^3 \\ \hline & \quad x^3 + 2 \\ & \quad x^3 - x \\ \hline & \qquad x + 2 \end{array}$$

Therefore $\dfrac{x^5+2}{x^2-1} = x^3 + x + \dfrac{x+2}{x^2-1}$

By expressing $\dfrac{x+2}{x^2-1} = \dfrac{x+2}{(x-1)(x+1)}$ as partial fractions, we obtain

$$\frac{x^5+2}{x^2-1} = x^3 + x - \frac{1}{2(x+1)} + \frac{3}{2(x-1)}$$

Using partial fractions for integration

Remember, if the degree of the numerator is greater than or equal to the degree of the denominator (improper), then division must take place first.

Example-17 (from **Example 16**)

Find $\displaystyle\int \frac{x^5+2}{x^2-1}\,dx.$

Expressing as partial fractions:

$$\frac{x^5+2}{x^2-1} = x^3 + x - \frac{1}{2(x+1)} + \frac{3}{2(x-1)}$$

Hence

$$\begin{aligned}\int \frac{x^5+2}{x^2-1}\,dx &= \int x^3 + x - \frac{1}{2(x+1)} + \frac{3}{2(x-1)}\,dx \\ &= \frac{x^4}{4} + \frac{x^2}{2} - \frac{1}{2}\ \ln|x+1| + \frac{3}{2}\ \ln|x-1| + c \\ &= \frac{x^4}{4} + \frac{x^2}{2} + \frac{1}{2}\ \ln\left(\frac{|x-1|^3}{|x+1|}\right) + c\end{aligned}$$

Example-18

Express $\dfrac{3x+1}{(x+2)^2}$ in partial fractions and hence find $\displaystyle\int \frac{3x+1}{(x+2)^2}\,dx.$

Write $\dfrac{3x+1}{(x+2)^2} = \dfrac{A}{x+2} + \dfrac{B}{(x+2)^2}$

Then $3x+1 = A(x+2) + B$

Substituting $x = -2$ gives $-5 = B$.

Substituting $x = 0$ gives $1 = 2A + B$ and therefore $A = 3$.

$$\begin{aligned}\therefore \quad \int \frac{3x+1}{(x+2)^2}\,dx &= \int \frac{3}{x+2} - \frac{5}{(x+2)^2}\,dx \\ &= 3\ \ln|x+2| + \frac{5}{x+2} + c\end{aligned}$$

Example-19

Find an antiderivative of $\dfrac{4}{(x+1)(x^2+1)}$ by first expressing it as partial fractions.

$$\frac{4}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

Then $\quad 4 = A(x^2+1) + (Bx+C)(x+1)$

Let $x = -1$: $\quad 4 = 2A \quad \therefore A = 2$

Let $x = 0$: $\quad 4 = A + C \quad \therefore C = 2$

Let $x = 1$: $\quad 4 = 2A + 2(B+C) \quad \therefore B = -2$

We now turn to the integration:

$$\int \frac{4}{(x+1)(x^2+1)}\,dx = \int \frac{2}{x+1} + \frac{2-2x}{x^2+1}\,dx$$

$$= \int \frac{2}{x+1}\,dx + \int \frac{2}{x^2+1}\,dx - \int \frac{2x}{x^2+1}\,dx$$

$$= 2\ln|x+1| + 2\arctan x - \ln(x^2+1) + c$$

$$= \ln\left(\frac{(x+1)^2}{x^2+1}\right) + 2\arctan x + c$$

F. Approximation integration: trapezoidal and Simpson's rules

1) Some functions simply do not have antiderivatives that are elementary functions (e.g. $\sqrt{x}\cos x$, $\sin x^2$, $\sqrt{1-x^3}$). Numerical integration can be used to find the definite integral when finding an antiderivative is not feasible. Two numerical integration techniques are described in this section.

2) **The Trapezoidal Rule**

Let f be continuous on $[a, b]$. The Trapezoidal Rule for approximating $\int_a^b f(x)\,dx$ is

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2n}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right].$$

Moreover, as $n \to \infty$, the right-hand side approaches $\int_a^b f(x)\,dx$.

3) **Simpson's Rule**

Let f be continuous on $[a, b]$ and let n be an even integer. Simpson's Rule for approximating $\int_a^b f(x)\,dx$ is

$$\int_a^b f(x)\,dx \approx \frac{b-a}{3n}\left[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)\right].$$

Moreover, as $n \to \infty$, the right-hand side approaches $\int_a^b f(x)\,dx$.

4) Observe that the coefficients in Simpson's Rule have the following pattern.

1 4 2 4 2 4 . . . 4 2 4 1

And, n must be an even integer.

Example-20

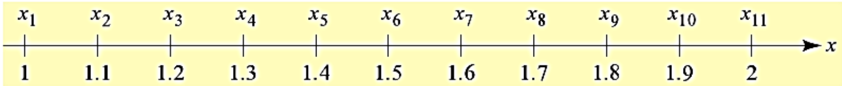
Use the trapezoidal rule with $n = 10$ to approximate $\int_1^2 \frac{1}{x}\,dx$.

Since $\Delta x = \dfrac{2-1}{10} = 0.1$

the interval $1 \le x \le 2$ is divided into 10 subintervals by the points

$$x_1 = 1,\ x_2 = 1.1,\ x_3 = 1.2,\ \ldots,\ x_{10} = 1.9,\ x_{11} = 2$$

as shown



Then, by the trapezoidal rule,

$$\int_1^2 \frac{1}{x}\,dx \approx \frac{0.1}{2}\left(\frac{1}{1} + \frac{2}{1.1} + \frac{2}{1.2} + \frac{2}{1.3} + \frac{2}{1.4} + \frac{2}{1.5} + \frac{2}{1.6} + \frac{2}{1.7} + \frac{2}{1.8} + \frac{2}{1.9} + \frac{1}{2}\right)$$
$$\approx 0.693771$$

We may evaluate it directly

$$\int_1^2 \frac{1}{x}\,dx = \ln|x|\Big|_1^2 = \ln 2 \approx 0.693147$$

Thus, the approximation of this particular integral by the trapezoidal rule with $n = 10$ is accurate (after round off) to two decimal places.

Example-21

Use Simpson's Rule to approximate $\int_0^{\pi} \sin x\,dx$. Compare the results for $n = 4$ and $n = 8$.

When $n = 4$, you have

$$\int_0^{\pi} \sin x\,dx \approx \frac{\pi}{12}\left(\sin 0 + 4\sin\frac{\pi}{4} + 2\sin\frac{\pi}{2} + 4\sin\frac{3\pi}{4} + \sin\pi\right) \approx 2.005.$$

When $n = 8$, you have $\displaystyle\int_0^{\pi} \sin x\,dx \approx 2.0003.$

G. Error bound for trapezoidal and Simpson's rule

1) Errors in the Trapezoidal Rule and Simpson's Rule

If f has a continuous second derivative on $[a, b]$, then the error E in approximating $\int_a^b f(x)\,dx$ by the Trapezoidal Rule is

$$|E| \le \frac{(b-a)^3}{12n^2}\left[\max|f''(x)|\right], \quad a \le x \le b. \qquad \text{Trapezoidal Rule}$$

Moreover, if f has a continuous fourth derivative on $[a, b]$, then the error E in approximating $\int_a^b f(x)\,dx$ by Simpson's Rule is

$$|E| \le \frac{(b-a)^5}{180n^4}\left[\max|f^{(4)}(x)|\right], \quad a \le x \le b. \qquad \text{Simpson's Rule}$$

2) The theorem above states that the errors generated by the Trapezoidal Rule and Simpson's Rule have upper bounds dependent on the extreme values of $f''(x)$ and $f^{(4)}(x)$ in the interval $[a, b]$. Furthermore, these errors can be made arbitrarily small by increasing n, provide that f'' and $f^{(4)}$ are continuous and therefore bounded in $[a, b]$.

3) For Simpson's rule, the upper bound on the absolute error is proportional to $(\Delta x)^4$, whereas the upper bound on the absolute error for the trapezoidal approximations is proportional to $(\Delta x)^2$. Thus, reducing the interval width by a factor of 10, for example, reduces the error bound by a factor of 100 for the trapezoidal approximations but reduces the error bound by a factor of 10,000 for Simpson's rule. This suggests that, as n increases, the accuracy of Simpson's rule improves much more rapidly than that of the trapezoidal approximations.

Example-22

Estimate the accuracy of the approximation of $\int_1^2 \frac{1}{x}dx$ by the trapezoidal rule with $n = 10$.

Starting with $f(x) = \frac{1}{x}$, compute the derivatives

$$f'(x) = -\frac{1}{x^2} \qquad \text{and} \qquad f''(x) = \frac{2}{x^3}$$

and observe that the largest value of $|f''(x)|$ for $1 \le x \le 2$ is $|f''(1)| = 2$.

Apply the error formula with

$$M = 2 \qquad a = 1 \qquad b = 2 \qquad \text{and} \qquad n = 10$$

to get $$|E_{10}| \le \frac{2(2-1)^3}{12(10)^2} \approx 0.00167$$

That is, the error in the approximation is guaranteed to be no greater than 0.00167.

In fact, to five decimal places, the error is 0.00062, as you can see: $0.693771 - \ln 2 = 0.00624$

Example-23

At least how many subintervals should be used in approximating $\ln 2 = \int_1^2 \frac{1}{x}dx$ by Simpson's rule for five decimal-place accuracy?

Recognize $$f(x) = \frac{1}{x}, \quad a = 1, \quad b = 2$$

Differentiate $$f'(x) = -\frac{1}{x^2}, \quad f''(x) = \frac{2}{x^3}, \quad f'''(x) = -\frac{6}{x^4}, \quad f^{(4)}(x) = \frac{24}{x^5}$$

$$\left|f^{(4)}(x)\right| = \left|\frac{24}{x^5}\right| = \frac{24}{x^5}$$

For $x \in [1,2]$, $K_4 = \max\left|f^{(4)}(x)\right| = 24$

To obtain five decimal-place accuracy, we must choose the number of subintervals so that $|E_S| \le 5 \times 10^{-6}$
This can be achieved by taking n in Simpson's rule to satisfy

$$\frac{(b-a)^5 K_4}{180\, n^4} \le 5 \times 10^{-6}$$

$$\frac{1^5 \cdot 24}{180 \cdot n^4} \le 5 \times 10^{-6}$$

$$n^4 \ge \frac{2 \times 10^6}{75}$$

$$n \ge \approx 12.779$$

Since <u>n must be an even integer</u>, the smallest value of n that satisfies this requirement is $n = 14$. Thus, the approximation using 14 subintervals will produce five decimal-place accuracy.

Note: to achieve m decimal-place accuracy, the error should be $\leqslant 5 \times 10^{-(m+1)}$.